10. $\lim_{x\to\frac{\pi}{2}}(\sin x)^{\tan x}$ 11. $\lim_{x\to 0^+}(\cos x)^{\frac{1}{x^2}}$

12. If an initial amount A_0 of money is invested at an interest rate r compounded n times a year, the value of the investment after t years is $A = A_0\left(1+\frac{r}{n}\right)^{nt}$. If the interest is compounded continuously, (that is as $n\to\infty$), show that the amount after t years is $A = A_0e^{rt}$

7.6 Applications of First Derivative

Using the first derivative we can test a function $f(x)$ for its monotonicity (increasing or decreasing), focusing on a particular point in its domain and the local extrema (maxima or minima) on a domain.

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7.6.1 Monotonicity of functions

Monotonicity of functions are its behaviour of increasing or decreasing.

Definition 7.4

A function $f(x)$ is said to be an increasing function in an interval I if $a<b \Rightarrow f(a)\le f(b), \forall a,b\in I$.

Definition 7.5

A function $f(x)$ is said to be a decreasing function in an interval I if $a<b \Rightarrow f(a)\ge f(b), \forall a,b\in I$.

The function $f(x)=x$ is an increasing function in the entire real line, whereas the function $f(x)=-x$ is a decreasing function in the entire real line. In general, a given function may be increasing in some interval and decreasing in some other interval, say for instance, the function $f(x)=|x|$ is decreasing in $(-\infty,0]$ and is increasing in $[0,\infty)$. These functions are simple to observe for their monotonicity. But given an arbitrary function how we determine its monotonicity in an interval of a real line? That is where following theorem (stated without proof) will be useful.

Theorem 7.7

If the function $f(x)$ is differentiable in an open interval (a,b) then we say,

(1) if

$$\frac{d}{dx}(f(x)) \ge 0, \forall x\in(a,b), \quad \text{... (1)}$$

then $f(x)$ is increasing in the interval (a,b),

(2) if

$$\frac{d}{dx}(f(x)) > 0, \forall x\in(a,b), \quad \text{... (2)}$$

then $f(x)$ is strictly increasing in the interval (a,b).

The proof of the above can be observed from Theorem 7.3.

(3) $f(x)$ is decreasing in the interval (a,b) if

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